

Honeywell Campus Connect Hackathon

- **Problem Statement ID - 1**
- **Problem Statement Title - Eco-Loop Building Agents**
- **Theme - Smart Building**
- **PS Category - MS**
- **Student Name - Kumar Ritesh Raushan**
- **Student ID**



Energent

AI-Powered Building Energy Optimization Platform

Problem:

- Buildings consume large amounts of energy.
- Traditional BMS uses fixed schedules.
- Engineers manually analyze simulation reports.
- Optimization takes significant time and effort.

Solution - Energent:

- AI-powered autonomous optimization platform.
- Uses EnergyPlus for building simulation.
- AI Agent analyzes simulation results.
- Automatically modifies building parameters.
- Repeats optimization until the best result is achieved.



Technical Approach

➤ **Frontend** - React, Tailwind CSS, Recharts

➤ **Backend** - FastAPI, Python

➤ **AI** - OpenSource LLM, MCP Tool Calling

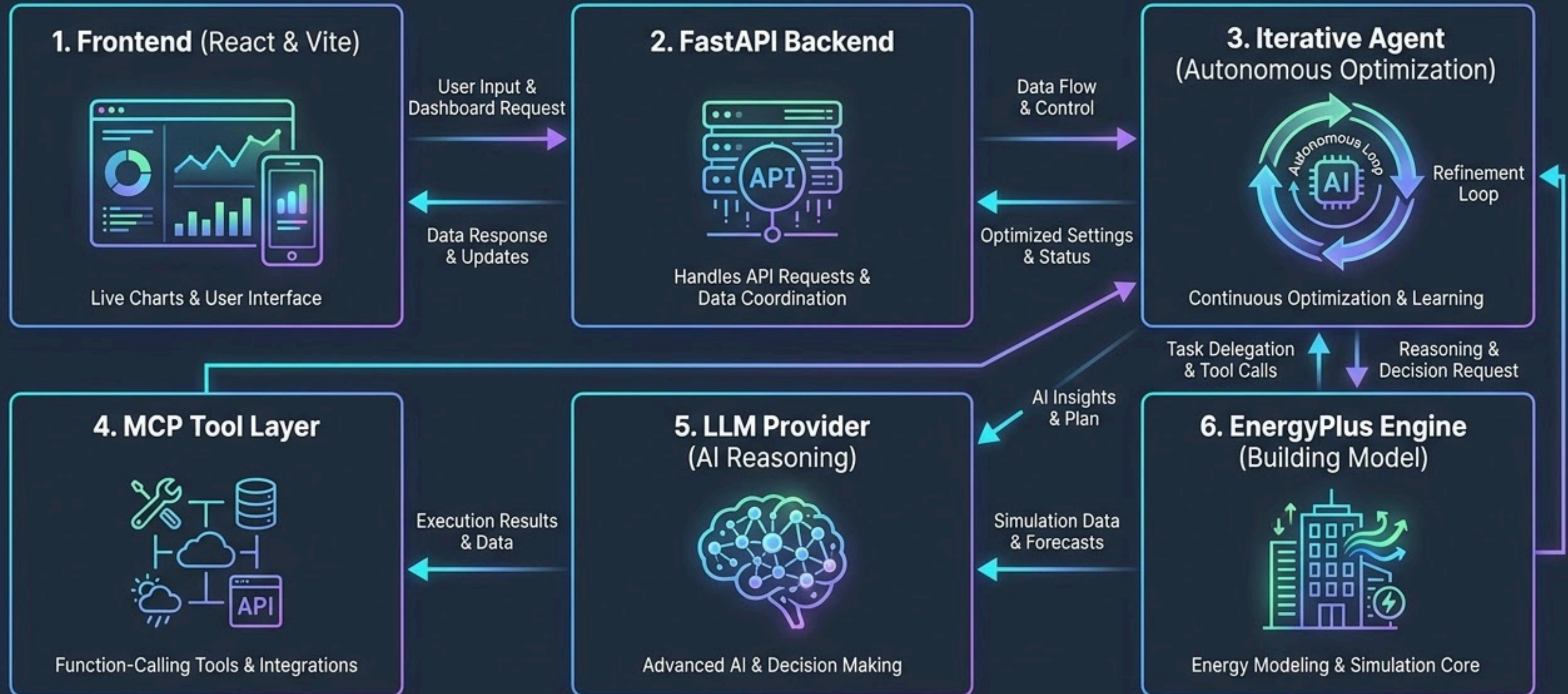
➤ **MCP Tools** - Parse IDF, Run Simulation, Analyze Results, Generate ECMs, Modify Setpoints

➤ **Working Process** -

1. The user starts a building simulation from the web dashboard.
2. The FastAPI backend launches EnergyPlus to simulate the building.
3. Simulation results are continuously streamed to the dashboard using Server-Sent Events.
4. The AI Agent analyzes energy consumption, thermal comfort, and building performance.
5. Using MCP tool calling, the AI automatically selects the appropriate tools to perform different tasks.
6. The building model is modified with optimized parameters and the simulation is executed again.
7. The optimization continues until energy savings stop improving while maintaining occupant comfort.

Technical Approach

Energent System Architecture



Innovation & Feasibility.

➤ **Autonomous AI Optimization -**

Unlike traditional simulation tools, Energent does not stop after generating results. The AI Agent continuously improves the building model by performing multiple optimization rounds automatically.

➤ **MCP Tool Calling** - The AI interacts with the system using specialized tools such as:

- Parse Building Model
- Run Simulation
- Analyze Results
- Generate Energy Conservation Measures
- Modify Building Parameters
- Detect Errors

This allows the AI to make intelligent decisions instead of generating only text responses.

➤ **Feasibility -**

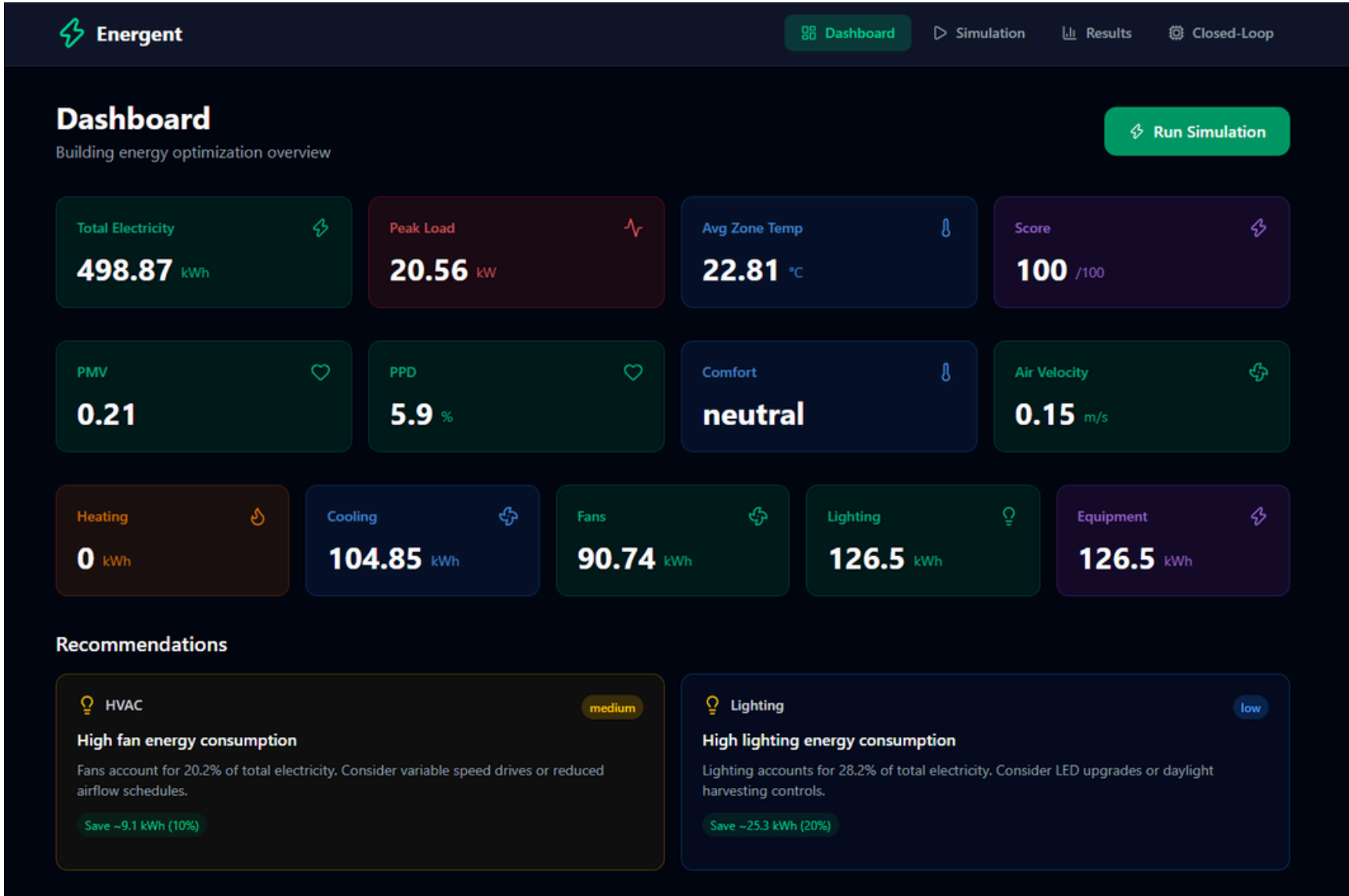
- Our platform evaluates occupant comfort using internationally accepted PMV and PPD indices based on the ASHRAE 55 standard.
- This ensures that energy savings are achieved without compromising indoor comfort.

➤ **Feasibility -**

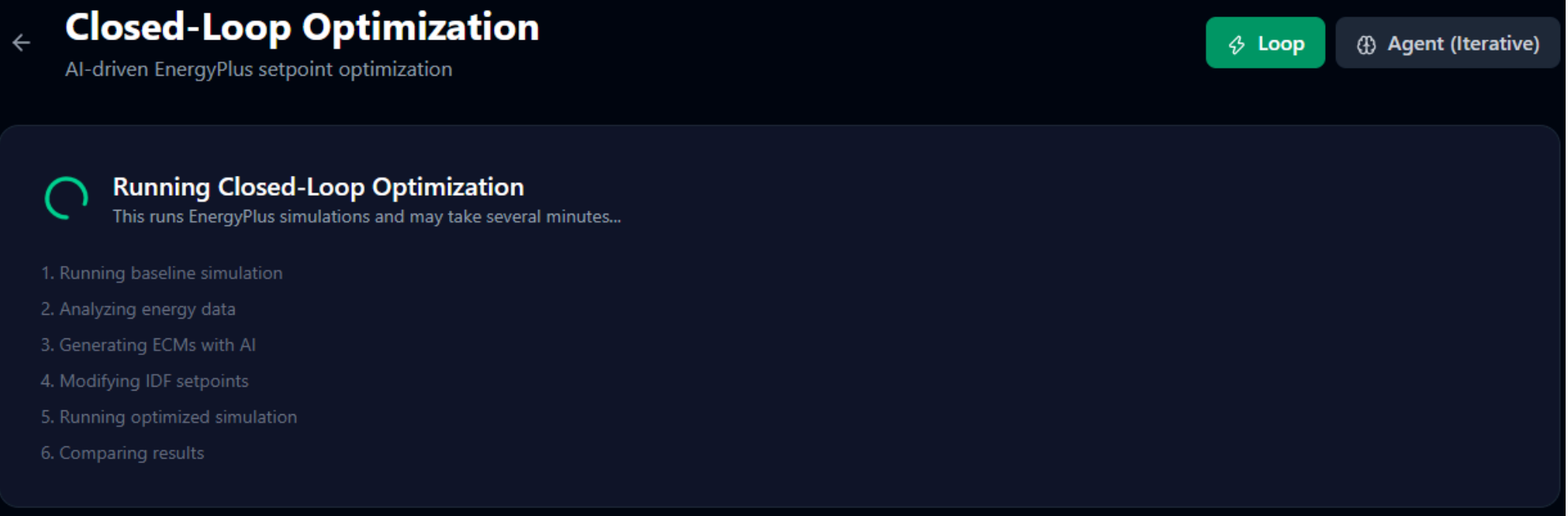
- The solution is practical because it is built using industry-standard technologies including EnergyPlus, FastAPI, React, and open-source Large Language Models.
- The architecture can also be extended to integrate with IoT devices and real Building Management Systems.

Results & Artifacts

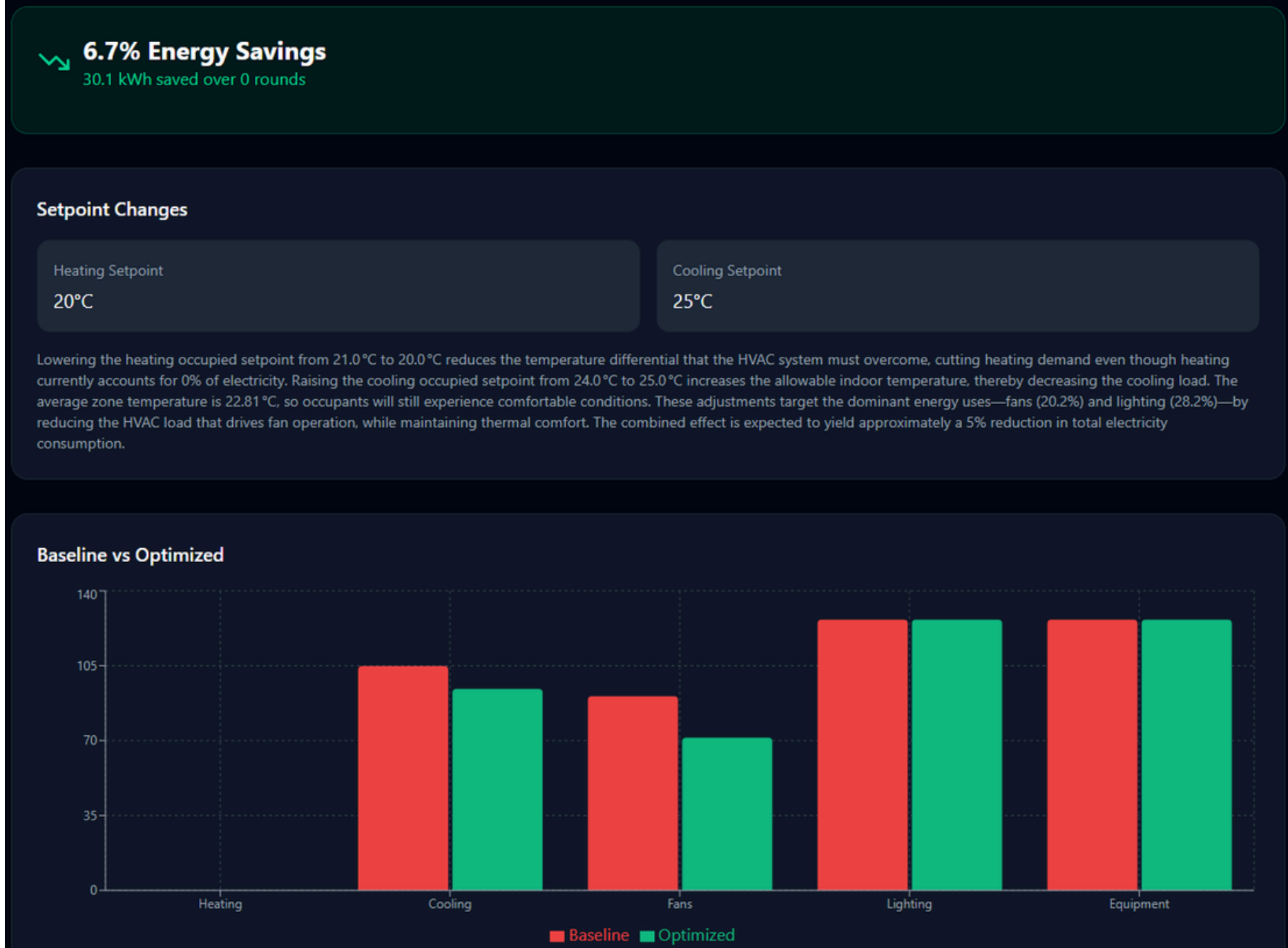
Dashboard



Results



Results & Artifacts



Conclusion

- Energent combines building simulation, artificial intelligence, and autonomous decision-making into a single platform.
- By automatically analyzing simulation results and optimizing building parameters, it helps reduce energy consumption, improve occupant comfort, and minimize the manual effort required in traditional building optimization workflows.

Traditional Approach	Energent
• Manual report analysis • Engineer changes building model • Single simulation • Static reports • Manual trial-and-error	• AI automatically analyzes results • AI modifies the building model • Multi-round autonomous optimization • Real-time streaming dashboard • Intelligent recommendations with thermal comfort analysis